OnionGuard

AI-Powered Onion Disease Detection for Farmers

Nkabom Honours Team | University of Ghana | 2026

Currently Live on Google Play Store

Meet the Team

Nkabom Honours Team | University of Ghana | 2026

Elvis Asare Nkrumah

Department of Computer Science and Statistics
University of Ghana

Emmanuel Bobbie

Department of Crop Science
University of Ghana

Issah Jarah Muftawu

Department of Materials Science and Engineering
University of Ghana

Benedicta Afi Agbodzi

Team Lead

Department of Communication Studies (MPhil)
University of Ghana

Nathaniel Agbesi Adika

Department of Computer Engineering
University of Ghana

The Problem

Why OnionGuard is needed

50%

Crop losses in West Africa due to pests & diseases

ZERO

Pre-existing ML models for onion disease classification

Key Challenges

- No ML models exist for the TOM2024 onion disease dataset
- Limited access to agricultural extension officers in rural areas
- Language barriers prevent farmers from accessing disease information
- No affordable diagnostic tools for smallholder farmers
- No centralized system for tracking regional disease outbreaks

Source: Appiah et al., 2025 (Data in Brief, Elsevier)

Dataset — TOM2024

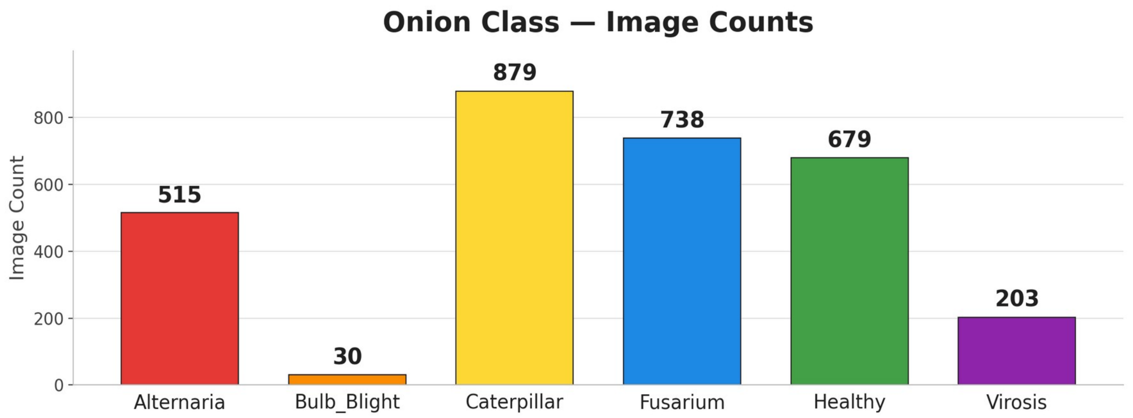
Maize, Tomato & Onion Multi-class Data

About TOM2024

- Published in Data in Brief (Elsevier), 2025
- Field-collected from farms in Burkina Faso
- Covers healthy crops, pests and diseases
- Addresses the West-African ML data gap
- First public benchmark for onion classes

Onion Subset — what we used

- 6 classes: 1 healthy + 4 diseases + 1 pest
- 3,044 onion images across all classes
- Resized from variable sizes to 224x224
- Imbalanced: Caterpillar 879 vs Bulb_Blight 30
- Class weights + 3-stage fine-tuning



Onion Classes

Six classes the model learns to distinguish

Healthy

Alternaria

Caterpillar

Virosis

Bulb Blight

Fusarium

Our Solution

OnionGuard mobile application

Disease Detection

On-device AI classifies 6 onion conditions from camera photos.

Freshness Check

Gemini Vision AI evaluates onion freshness

Treatment Guides

Detailed treatment in 5 languages with voice playback for low-literacy farmers.

Analytics

Personal & regional disease tracking dashboards for farmers and officers.

Security: JWT Auth + Bcrypt + Biometric Login + HTTPS (TLS 1.2/1.3) + Let's Encrypt SSL

Machine Learning Model

MobileNetV3-Large trained on TOM2024 dataset

Base: MobileNetV3-Large (ImageNet pretrained)

Quantization: Float16 for mobile deployment

Size: 5.72 MB (TFLite)

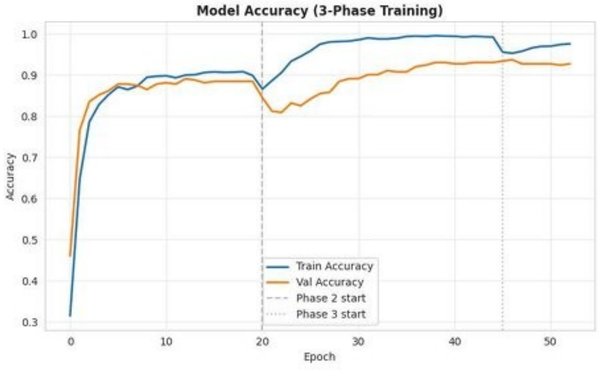
Part 3 — Model Training (3-Stage Progressive Fine-Tuning)

Stage	Layers Trained	Learning Rate	Epochs
Phase 1	Classifier head only	1e-3	15
Phase 2	Last ~30% of backbone + head	1e-4	20
Phase 3	All layers	1e-5	10

Optimizer: AdamW (weight_decay=0.01)

Loss: Categorical cross-entropy with label smoothing (0.1)

LR Schedule: Cosine annealing with 5-epoch linear warmup

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Best validation accuracy: 0.9375 (93.75%)
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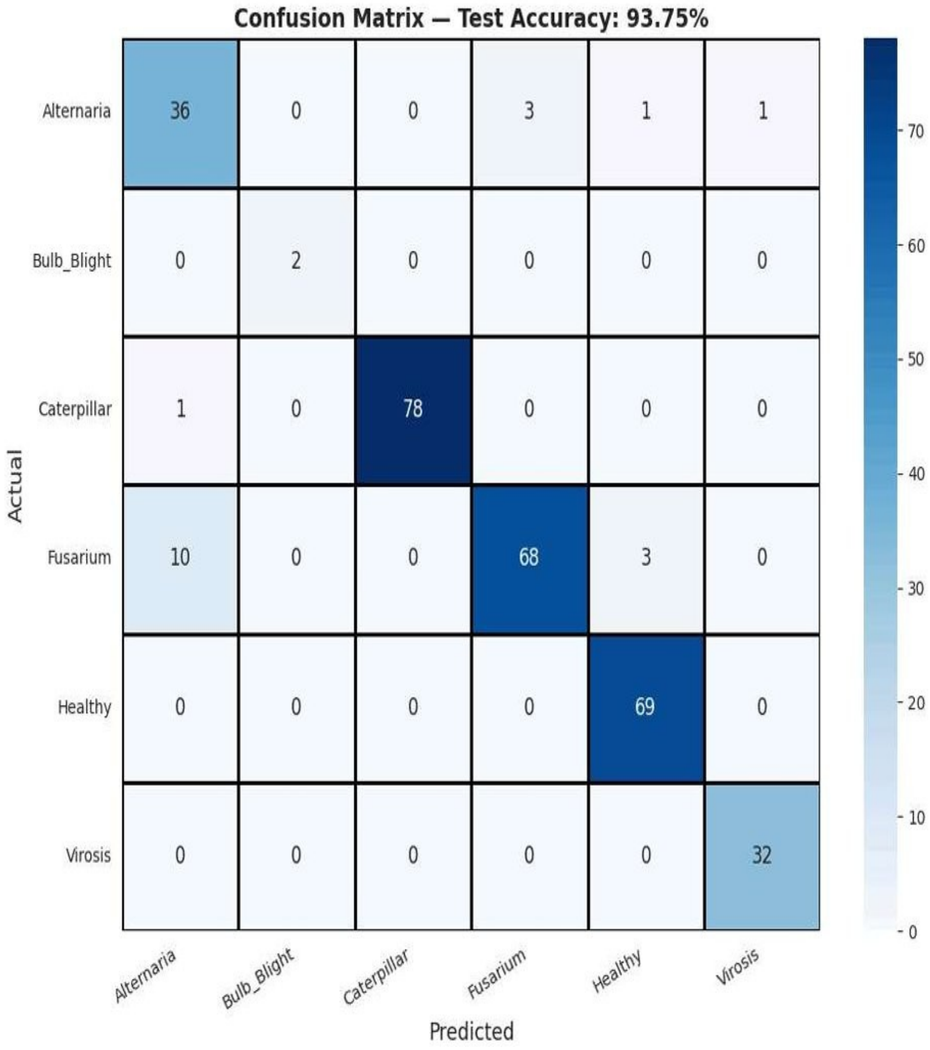


Model Performance

Evaluation on held-out test set (304 images)

93.75%

Overall Test Accuracy

TFLite: 93.75% | Size: 5.72 MB

Classification Report

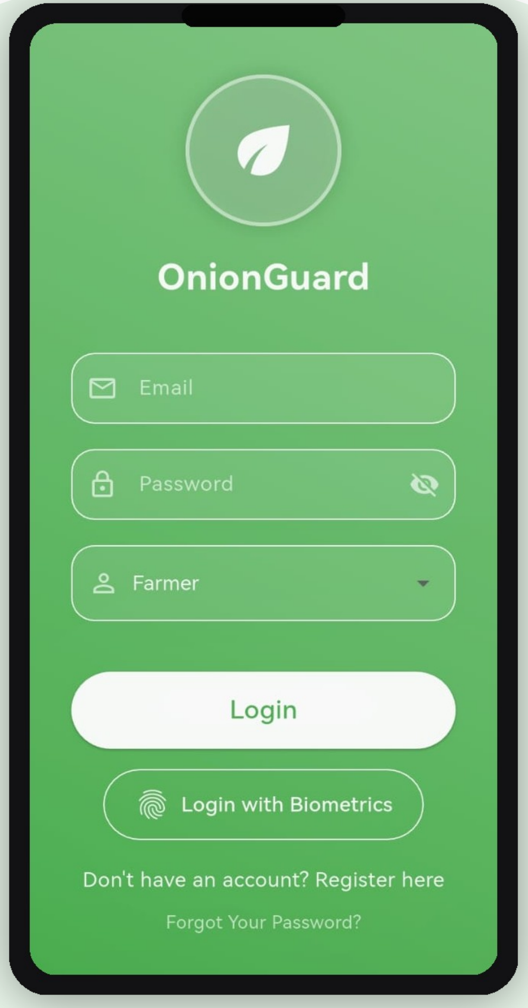
Class	Precision	Recall	F1-Score	Accuracy
Alternaria	0.766	0.878	0.818	87.80%
Bulb Blight	1.000	1.000	1.000	100.00%
Caterpillar	1.000	0.987	0.994	98.73%
Fusarium	0.958	0.840	0.895	83.95%
Healthy	0.945	1.000	0.972	100.00%
Virosis	0.970	1.000	0.985	100.00%
Macro Avg	**0.940**	**0.951**	**0.944**	

Key Insights

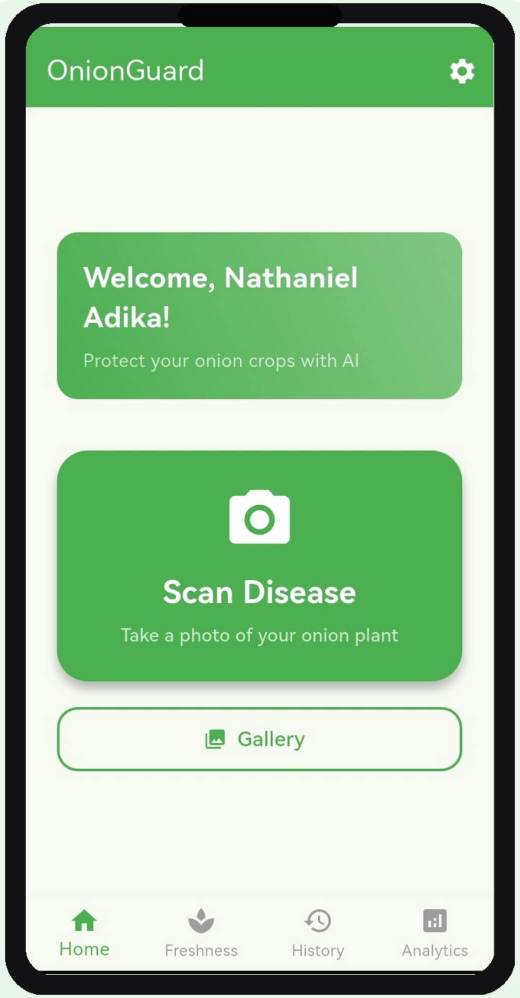
- Caterpillar, Healthy, Virosis & Bulb Blight achieve 98-100% accuracy
- Fusarium (83.95%) shows lowest accuracy due to visual similarity with Alternaria
- Class weights handle imbalance effectively (879 Caterpillar vs 30 Bulb Blight)
- TFLite quantization maintains full accuracy with 5.72 MB model size

Mobile Application (1/3)

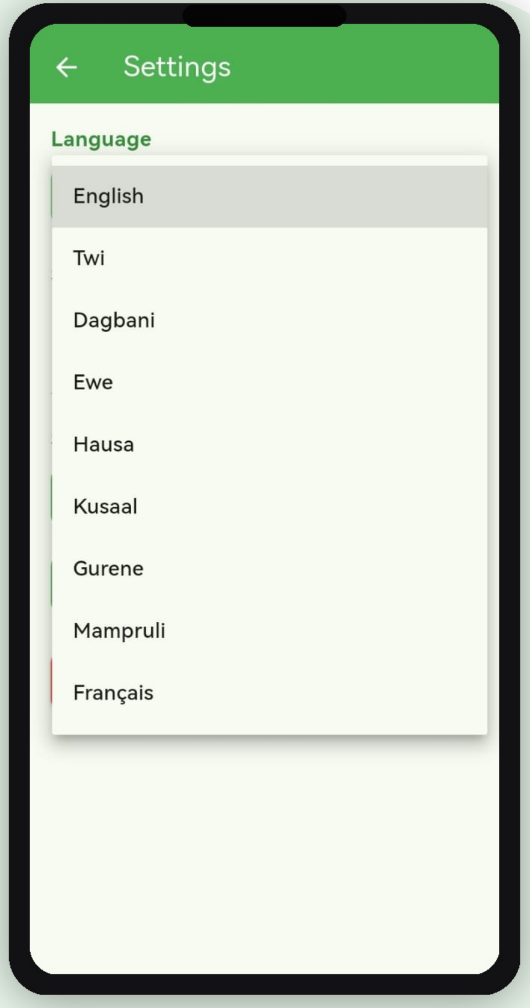
Fingerprint Login | Home Screen | 9 Languages



Fingerprint Login



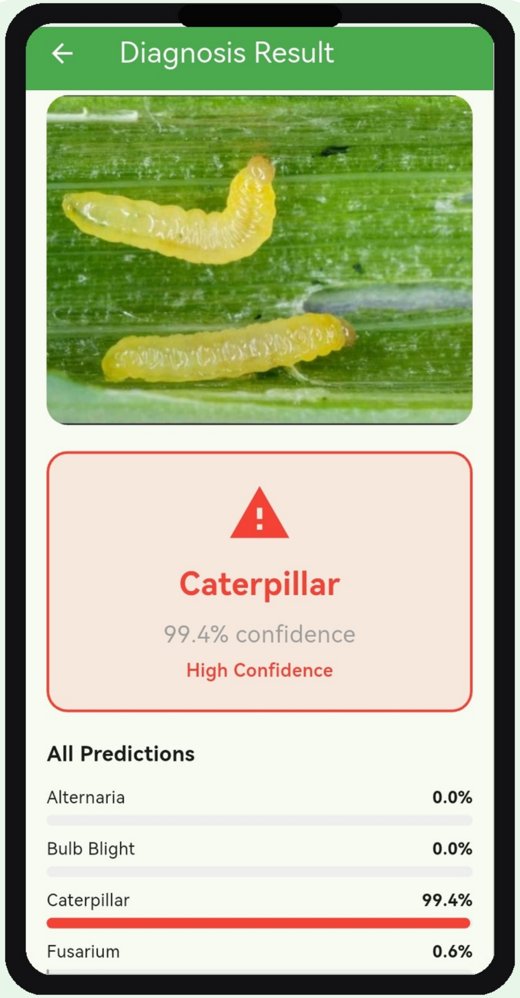
Home Screen



9 Languages

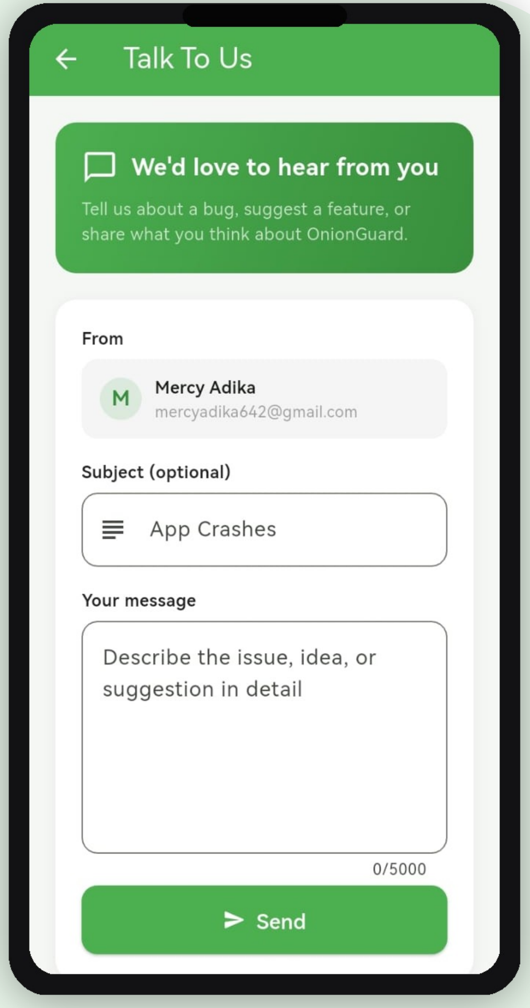
Mobile Application (2/3)

Disease Detection | Farm Analytics | Talk To Us



Disease Detection

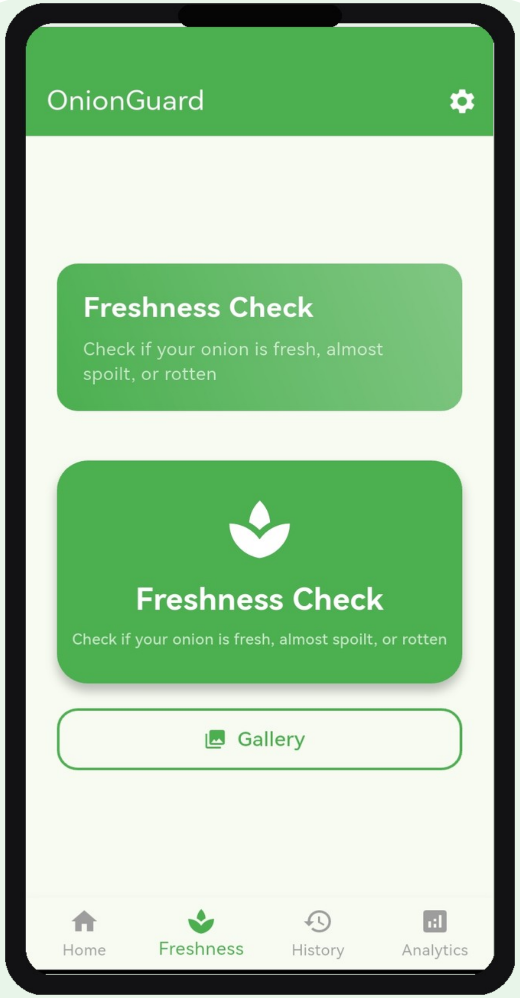
Farm Analytics



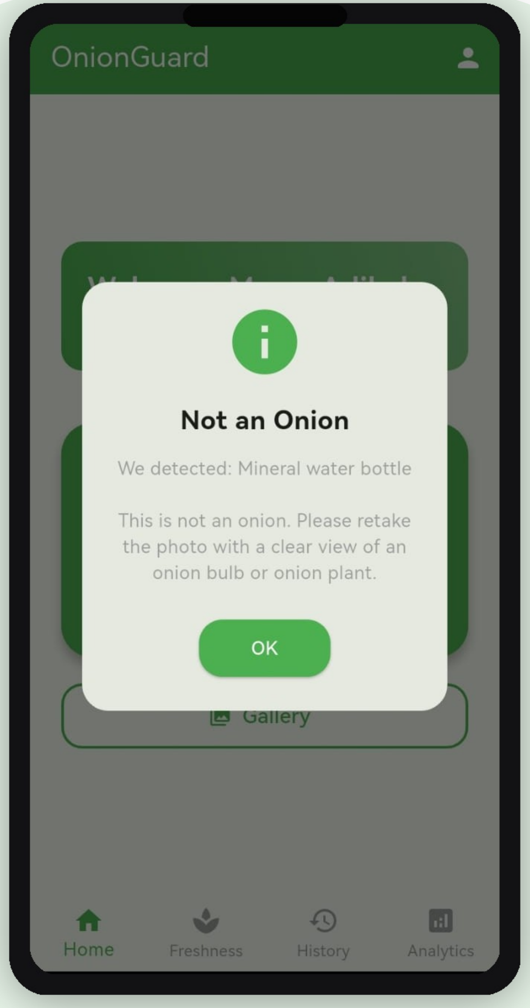
Talk To Us

Mobile Application (3/3)

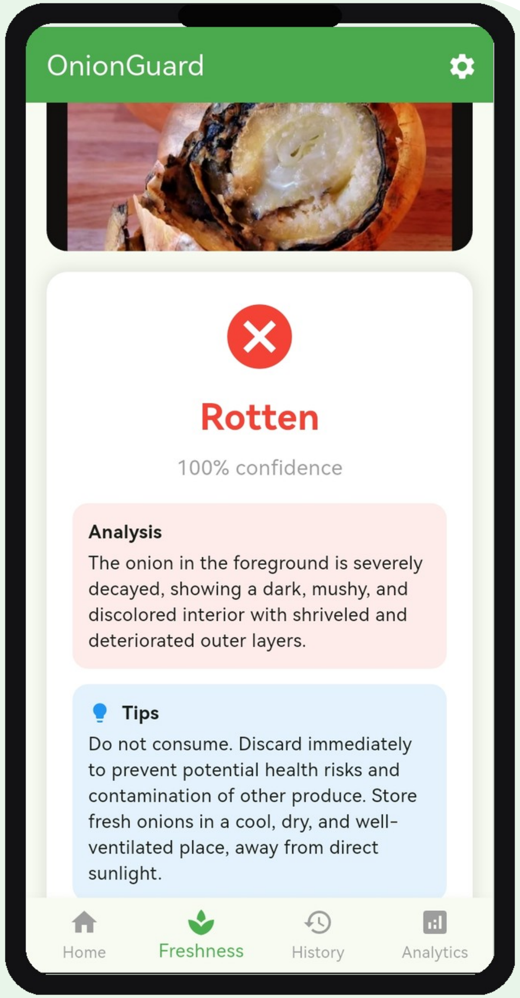
Freshness Check | Smart Pre-Check | AI Analysis



Freshness Check



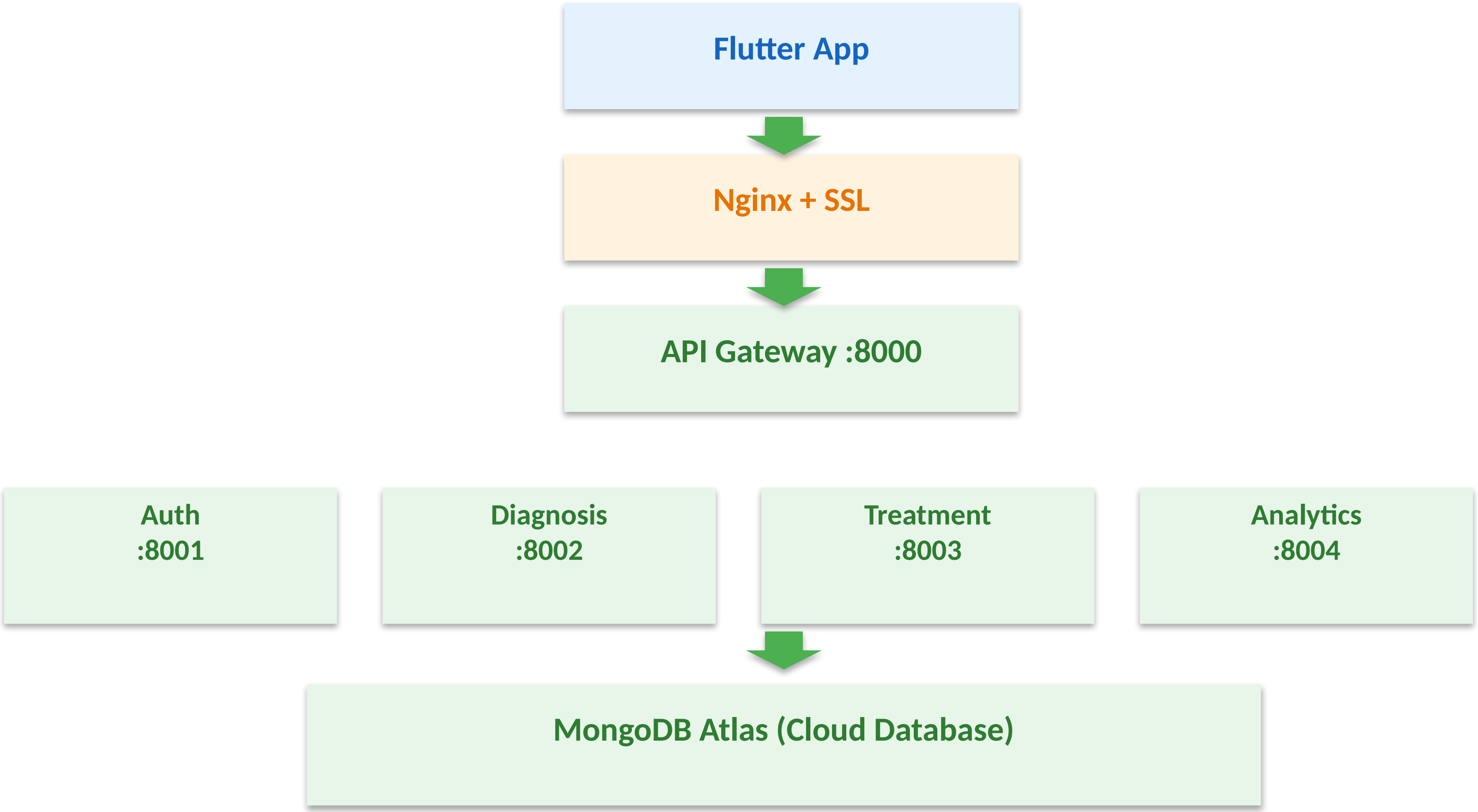
Smart Pre-Check



AI Analysis

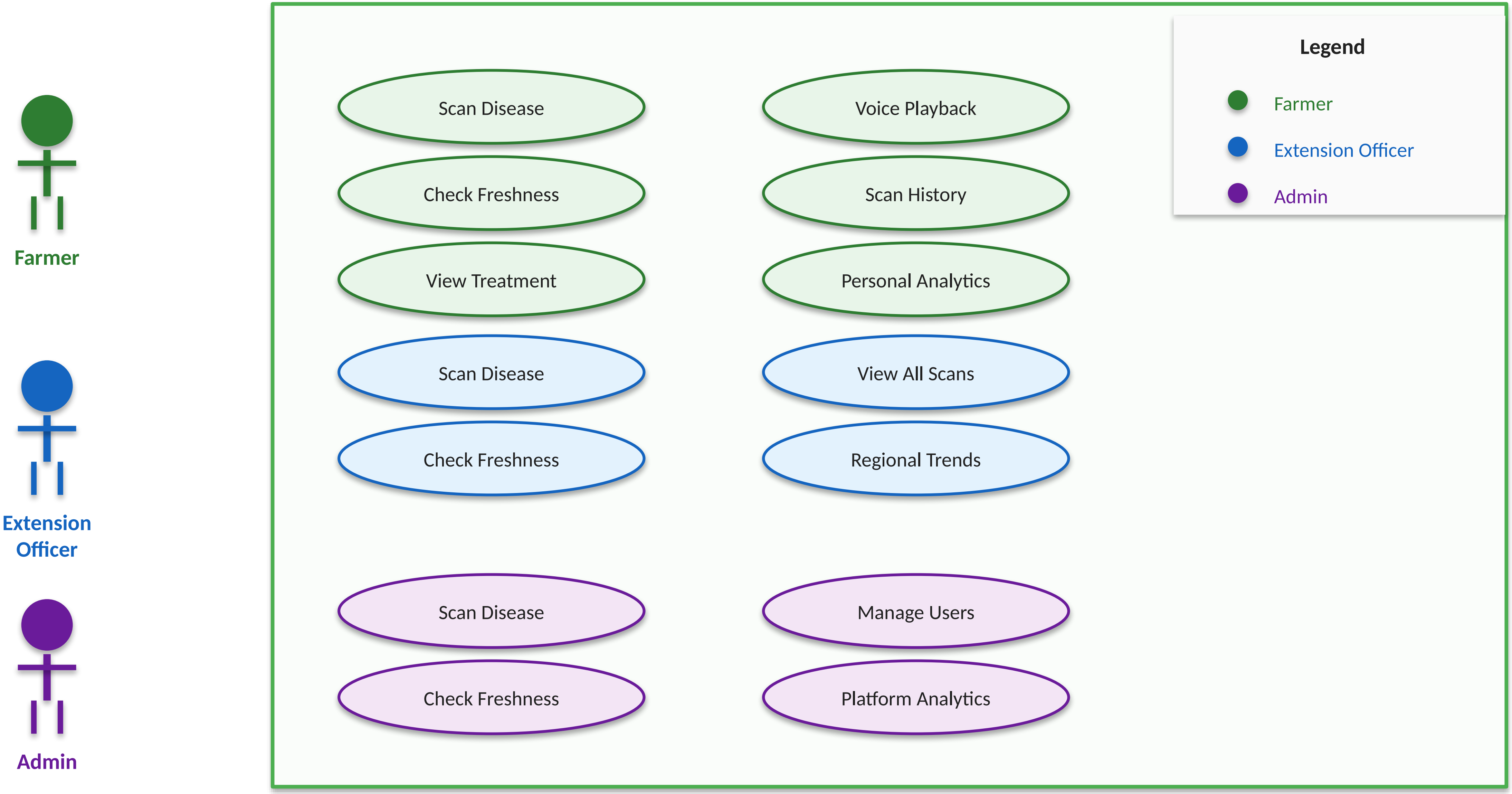
System Architecture

Microservices deployed on AWS EC2



Use Case Diagram

Role-based access control



Technology Stack

Full-stack modern architecture

Frontend	Flutter / Dart
Backend	FastAPI / Python 3.11
Database	MongoDB Atlas
ML Model	TensorFlow Lite
Vision AI	Google Gemini 2.5 Flash
Voice	Google TTS (gTTS)
State Mgmt	Provider

Charts	fl_chart
Auth	JWT + Bcrypt
Biometrics	local_auth
Containers	Docker + Compose
Proxy	Nginx
SSL	Let's Encrypt + Certbot
DNS	DuckDNS

Relevance of This Work

Why OnionGuard matters beyond the classroom

First ML Benchmark for Onion Disease

- Zero ML models existed for onion disease on TOM2024 before this project
- First production benchmark across 6 onion disease classes
- Directly closes gap identified by Appiah et al. (2025, Elsevier)

Food Security Impact in West Africa

- Up to 50% crop losses due to undetected pests and diseases
- Expert-level disease diagnosis in any farmer’s hands via smartphone
- Early intervention that protects yields and farm incomes at scale

Removes Language & Literacy Barriers

- 240+ translated UI strings across 9 local languages
- Google TTS voice playback for all treatment guides
- Reaches farmers excluded by English-only AgriTech tools

Human-in-the-Loop Feedback Pipeline

- Farmers flag wrong AI predictions directly in the app
- Admins export ML-ready correction ZIPs for model retraining
- Self-improving data flywheel grounded in real field conditions

Conclusion

OnionGuard: from research benchmark to live agricultural tool

Technical Achievement

- 93.75% test accuracy on held-out images
- 5.72 MB TFLite model runs fully on-device
- First ever ML benchmark for onion disease on TOM2024

Human Impact

- Smallholder farmers lose up to 50% of onion crops to disease
- Expert-grade diagnosis now accessible via any smartphone
- Culturally accessible: 9 languages with voice-guided treatment

Full-Stack Solution

- On-device AI + Gemini Vision freshness assessment
- Human-in-the-loop correction pipeline for continuous improvement
- Microservices on AWS EC2, designed to scale across West Africa

Path Forward

- Live on Google Play Store — beyond the academic prototype
- Field testing on onion farms across Ghana and West Africa
- Extend to other crops — Maize & Tomato (TOM2024) and other crops with prevalent diseases

References

Sources cited in this presentation

Dataset

- Appiah, M. K., et al. (2025). TOM2024: A multi-class image dataset for maize, tomato and onion disease and pest classification. *Data in Brief*, Elsevier. https://doi.org/10.1016/j.dib.2025.111413

ML Model

- Howard, A., et al. (2019). Searching for MobileNetV3. *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 1314–1324. https://doi.org/10.1109/ICCV.2019.00140

Project & Application

- Nkabom Honours Team. (2026). OnionGuard: AI-Powered Onion Disease Detection [Mobile Application]. University of Ghana. Available on Google Play Store. https://onion-guard.duckdns.org

Vision AI

- Google DeepMind. (2024). Gemini: A Family of Highly Capable Multimodal Models. Technical Report. https://doi.org/10.48550/arXiv.2312.11805

Open to Collaboration

Partner with us to expand OnionGuard across West Africa

https://onion-guard.duckdns.org

Thank You

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